

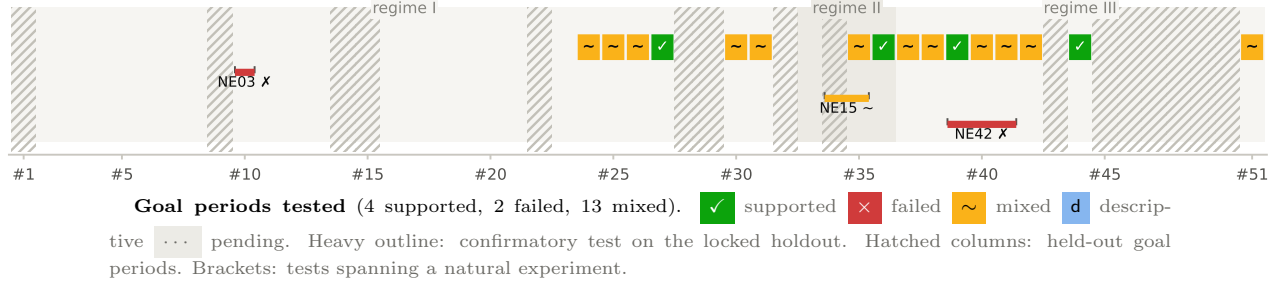
H18: Attention dilutes as 1/k: response to a message falls with the number of messages waiting at the agent's turn

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 48% Faithfulness 2.5/5 Usefulness 2/5

Bottom line. Correction (2026-10-04): the all-turn exponent ($\hat{\beta} \approx 0.66$) is not separable from reactive timing (that null gives 0.75–0.80); the identified dilution is #51's timer-wake batches, $k^{-0.45}$ [0.41, 0.50] (round 1b), where the null gives ≈ 0 . So coupling scales sub-linearly, not as a literal J/N ; replies go to the newest messages.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

At each talk turn of an agent, we count the k messages that reached it since its previous talk turn. We ask whether it responds to each pending sender, fitting within agent-days in 16 non-holdout goal periods. Round 1b recounts k on the context ledger (which model call received which message) and adds reply labels. Response falls with k in 16/16 periods: pooled $\hat{\beta} = 0.66 \pm 0.02$ (round 1: 0.63). The curve saturates or follows recency; a budget-like shape wins 6/16. In #51, timer-wake batches of exogenous size give $\hat{\beta} = 0.45$. Per-pair uptake falls with room size across periods ($\rho = -0.57$) and across #51's roster growth ($\rho \approx -0.8$). The round-1 placebo failure was mostly the visibility rule. Invisible senders now get addressed at half the pending rate (was 85%); conversation state explains the rest. The merge and fetch-limit tests fail. The holdout is not run.

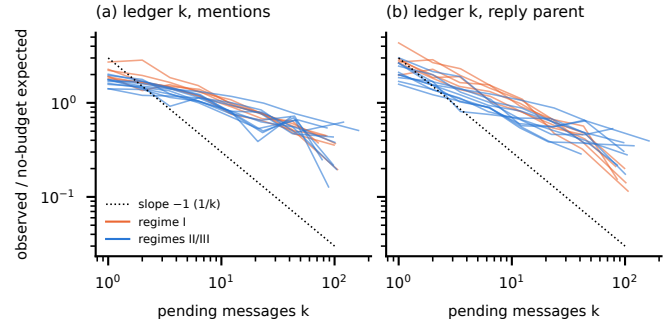
PHYSICAL MODEL

Mean-field coupling (01) with per-pair cross-excitation (09). At a talk turn of recipient i , the agent takes up each pending message m with hazard $\theta_{i,d} h_m$. It responds to sender j (a name or a reply parent) if it takes up any message of j . $\theta_{i,d}$: agent×day propensity. h : the attention rule (constant, $1/k$, saturating with k_0 , recency with ρ , power β). γ : boost for messages naming i . J_{ij} : per-pair coupling. N : room size. B : senders responded to per turn.

$$P(r_{\tau j} = 1) = 1 - \exp\left(-\theta_{i,d} \sum_{m \in P_j(\tau)} h_m\right)$$

$$h_m = e^{\gamma \mathbf{1}[m \rightarrow i]} \times \left\{1, \frac{1}{k}, \frac{1+k_0}{k_0+k}, \rho^{\text{rank}_m-1}, k^{-\beta}\right\}$$

$$J_{ij} \propto B k^{-\beta} \sim B N^{-\beta} \quad (\beta = 1: \text{literal } J/N)$$



Observables. Response relative to a no-budget model with the same agent×day propensities, vs pending k on the context ledger. One line per period. (a) Mentions: every period declines, less steeply than $1/k$ (dotted). (b) Reply parents: steeper, close to $1/k$, because each message has one parent. Recency decides which pending message gets the parent.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Expect per-pair engagement in a shared channel to fall as about $N^{-0.6}$ with room size N . Output per turn stays about the same, so a bigger room dilutes engagement and does not add collaboration. A reply goes to the newest pending messages. To raise within-room engagement, split the channel: the gain is about $(k \text{ ratio})^{0.6}$, as at the #35 split. This is a design rule, not a lever. It adds little to “use fewer, smaller rooms”, and the exponent varies by period (0.52–0.80). The mention signal is now cleaner, because the placebo failure was mostly a visibility bug. It still carries conversation state.

RESULTS VS PREDICTIONS

Prediction	Round 1 $\rightarrow$ round 1b (ledger k)	Verdict (1b)
P1 $\hat{\beta} > 0$, pooled 0.5–1.2	16/16, 0.63 $\rightarrow$ 16/16, 0.66 (replies 0.85)	yes
P2 budget shape beats recency	5/16 $\rightarrow$ 6/16; replies: 14/16	no
P4 mentions bypass dilution	1/16 $\rightarrow$ 1/16	no
P5 exogenous- k timer wakes	3/6 $\rightarrow$ 3/6; #51 0.50 $\rightarrow$ 0.45 [0.41, #51 yes 0.50]	yes
P6 room effect absorbed by k	3–4/8 $\rightarrow$ same	mixed
P7 05-04 merge (NE42 native)	failed $\rightarrow$ failed (raw); joint fit: k absorbs 53–86%	no
P8 regime invariance	Δ 0.14 $\rightarrow$ 0.07	yes
P9 $\bar{p}$ falls with N , $\hat{B}$ flat	ρ $-0.55/-0.03 \rightarrow -0.57/-0.02$	yes
#51 N sweep (native)	replies: $\bar{p}$ ρ -0.80 , $\hat{B}$ -0.64	mixed
P10 invisible placebo	0/16, $0.85\times$ pending $\rightarrow$ 3/16, mostly explained $0.49\times$	ex-

GOAL PERIODS IN DETAIL

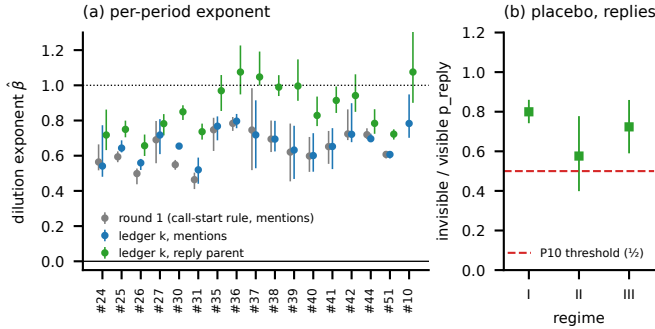
G24 $\sim$ 1b:	mixed	:	ledger k , mention response, pre-registere...
G25 $\sim$ 1b:	mixed	:	ledger k , mention response, pre-registere...
G26 $\sim$ 1b:	mixed	:	ledger k , mention response, pre-registere...
G27 $\checkmark$ 1b:	supported	:	ledger k , mention response, pre-regis...
G30 $\sim$ 1b:	mixed	:	ledger k , mention response, pre-registere...
G31 $\sim$ 1b:	mixed	:	ledger k , mention response, pre-registere...
G35 $\sim$ 1b:	mixed	:	ledger k , mention response, pre-registere...
G36 $\checkmark$ 1b:	supported	:	ledger k , mention response, pre-regis...
G37 $\sim$ 1b:	mixed	:	ledger k , mention response, pre-registere...
G38 $\sim$ 1b:	mixed	:	ledger k , mention response, pre-registere...

+9 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. (a) Exponent per period: round 1 (gray), ledger k with mentions (blue), ledger k with reply parents (green). The visibility fix barely moves it. Replies push it toward 1 by construction. (b) Placebo on reply labels: invisible pairs keep 58–80% of a visible pair’s reply score (threshold 50%).

CAVEATS

Mentions mix uptake with conversation state. Reply labels allow one parent per message and keep a thread-membership component. Neither response is a clean per-message uptake. Recency and budget remain inseparable where k is small. The timer-wake design identifies the exponent only in #51. Era confounds the cross-period size law, and #51 is the only large room. We apply no multiplicity correction. Nothing is confirmed on the holdout, and the holdout script still builds round-1 inputs.

ROUND 2

- **Essence.** Attention is the scarce resource, and coupling is its allocation.
- **H18-R1** A salience law, uptake = $f(\text{mention, recency, novelty, sender status})$, is fit and predicts held-out uptake (E5).
- **H18-R2** Total attention per turn is conserved; re-test with reply-threading labels.
- **H18-R3** An optimal room size maximizes total useful uptake (HH111).

H18-attention-dilution generated 2026-10-07 17:50 by infra/summaries/build_summaries.py card hypotheses/H18-attention-dilution/README.md ratings: H18 re-evaluation agent RE-V1 (suggested) (page 2)